

EXPERIMENT – 12

EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

AIM: To write an Embedded C program to multiply two 16-bit numbers using Keil software.

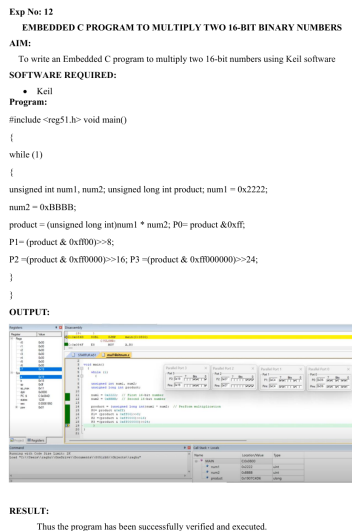
OBJECTIVES:

1. To define two 16-bit input numbers.
2. To perform 16-bit multiplication and obtain a 32-bit result.
3. To display the result through AT89C51 ports.

PROGRAM:

```
#include <reg51.h>
void main()
{
    while(1)
    {
        unsigned int num1, num2;
        unsigned long int product;
        num1 = 0x2222;
        num2 = 0xB BBB;
        product = (unsigned long int)num1 * num2;
        P0 = product & 0xFF;
        P1 = (product & 0xFF00) >> 8;
        P2 = (product & 0xFF0000) >> 16;
        P3 = (product & 0xFF000000) >> 24;
    }
}
```

OUTPUT IMAGE:



RESULT:

Thus the program has been successfully verified and executed.